

SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)
DepartmentofComputerScienceandEngineering

ASSESSMENTTOOL1-INTERACTIVETREESIMULATOR

CourseCode:	CSA0305	CourseTitle:	DataStructures
COAssessed:	CO3-Precision(BL3,BL5)	Assessment:	AssessmentTool1-Interactive Tree Simulator
Weightage:	30%of CIA	TotalMarks:	100
CO3Statement:	SolveproblemsforoptimizeddatastorageandretrievalusingBinarySearchTrees,AVLTrees,B-Trees, Red-Black Trees, and Splay Trees.		
StudentName:	K KABILAN	Reg. No.:	192521157
Section/Batch:	IT	Date:	29/07/2026

CodeofConduct

I K KABILAN (192521157) _____ (Name/RegNo)

certifythatthissubmissionismyoriginalworkandthatIhaveadheredtotheguidelinespecifiedforthis assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

SignatureoftheStudent:K.Kabilan

Instructions

- Thisassessmenttoolcontains5InteractiveTreeSimulatorquestions,Total:100Marks.
- Studentscanuseanyonline/offlineInteractiveTreeSimulator.Demonstrateinsertion,deletion, searching, and traversal operations wherever applicable.
- Whenastudentanswerusingsimulator,inevaluationtheanswersshouldhavecapturescreenshots after every major operation.
- SubmitthereportinPDFformatwithproperlylabelledscreenshotsasproofofcompletion.

Section/Topic		Focus	QNos	Marks
BinarySearchTree		Properties,binarytreeandbinarysearchtree,insertion and deletion	1	20
Tree Traversals		Traversaltypes,Inorder,Post orderandPreorder	2	20
AVLTree		Balancingfactor, Rotation,Typesofrotation,examples	3	20
BTree,2-3tree,2-3-4tree		Properties,structuresandOperations	4	20
RedBlackTree		Properties,ExampleandOperations	5	20
GRANDTOTAL:			100Marks	

Annexure A – Interactive Tree Simulator

- Construct a Binary Search Tree (BST) by inserting the following keys: 50, 30, 70, 20, 40, 60, 80. Perform:
 - Inorder Traversal
 - Search for 60
 - Delete 30
 - Display the final BST.
- Complete Tree Traversals by Inserting 50, 30, 70, 20, 40, 60, 80. Find all the traversals.
 - Inorder
 - Preorder
 - Postorder
- Insert 70, 50, 90, 40, 60, 80, 100, 55 into an AVL Tree. Delete 40 and 90. Analyze how the tree rebalances after each deletion.
- Construct a B-Tree of order 3 by inserting 45, 25, 65, 15, 35, 55, 75
- Insert 100, 80, 120, 60, 90, 110, 130 into a Red-Black Tree. Analyze their heights.

Rubrics for Calculation

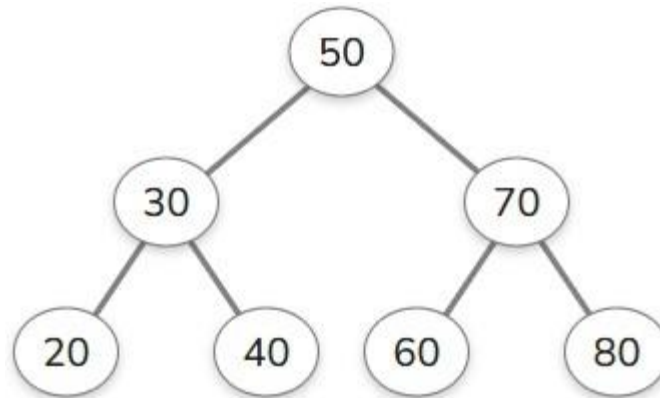
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Correct Tree Construction	20	Tree is constructed correctly with all operations performed accurately	Minor errors in construction	Some operations incorrect	Tree construction mostly incorrect
Understanding of Tree Operations	20	Clearly explains insertion, deletion, search and balancing operations	Explains most operations correctly	Limited understanding	Unable to explain operations
Analysis of Tree Behaviour	30	Correctly analyses balancing, rotations, nodes split/merge, splaying	Good analysis with minor omissions	Partial analysis	Poor or incorrect analysis
Presentation	20	Well-organized report with accurate observations and conclusion	Good report with minor issues	Basic report with limited observations	Incomplete report
Accuracy of Responses	10	90–100% correct answers	75–89% correct answers	50–74% correct answers	Below 50% correct answers

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

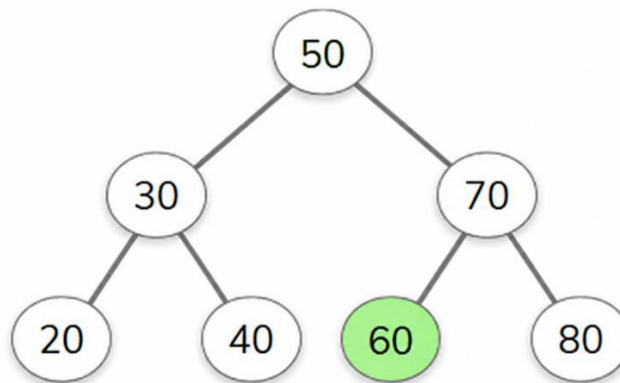
1. Construct a Binary Search Tree (BST) by inserting the following keys: 50, 30, 70, 20, 40, 60, 80

Perform:

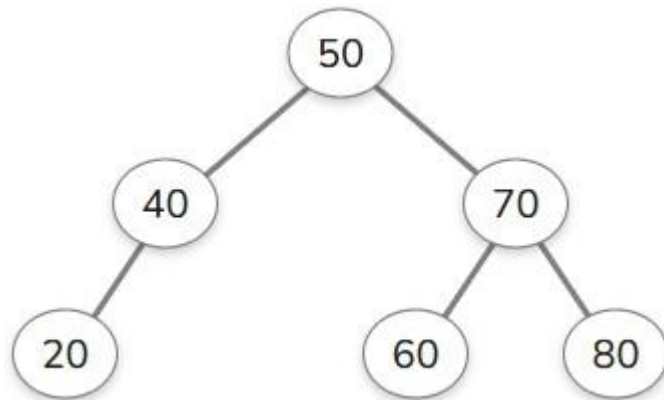
- a. **Inorder Traversal**



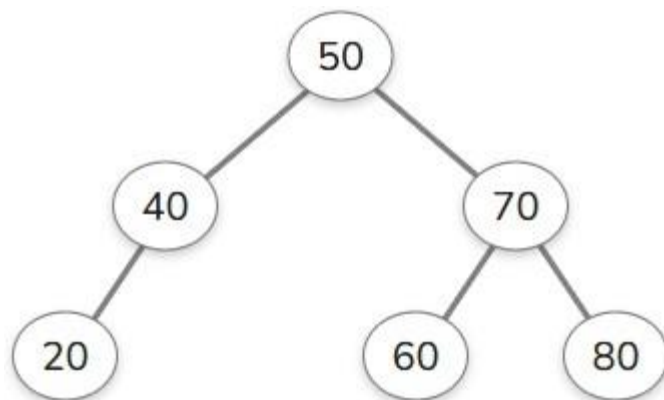
- b. **Search for 60**



c. Delete 30

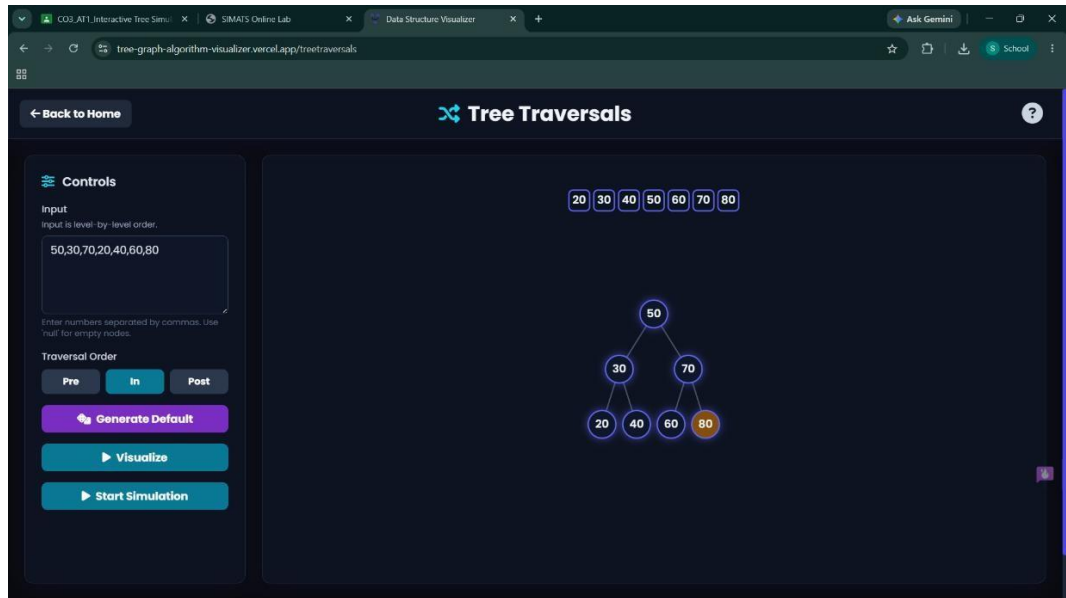


d. Display the final BST.

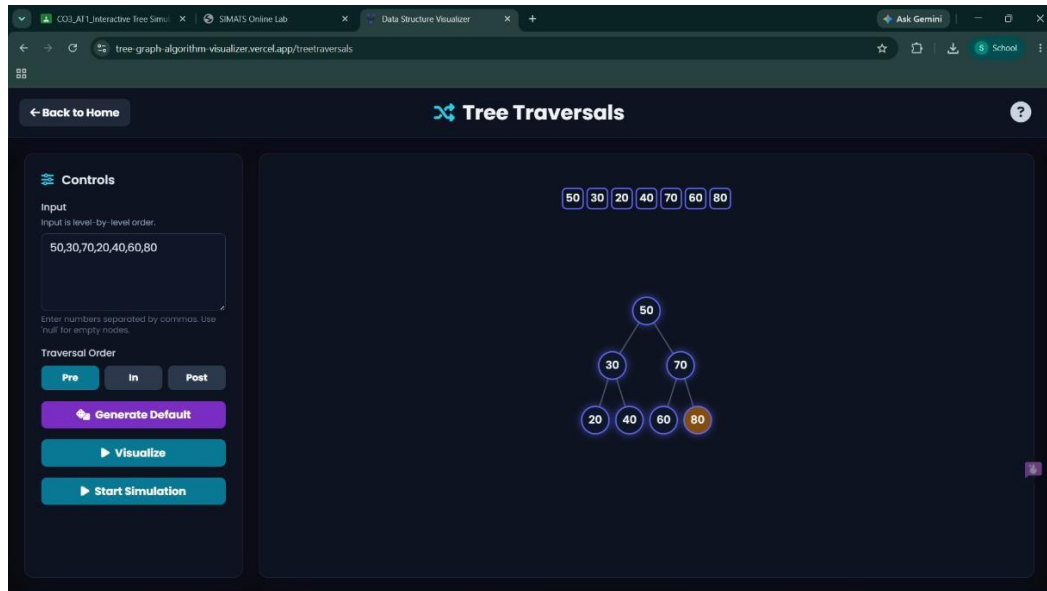


2. Complete Tree Traversals by Inserting 50, 30, 70, 20, 40, 60, 80. Find all the traversals.

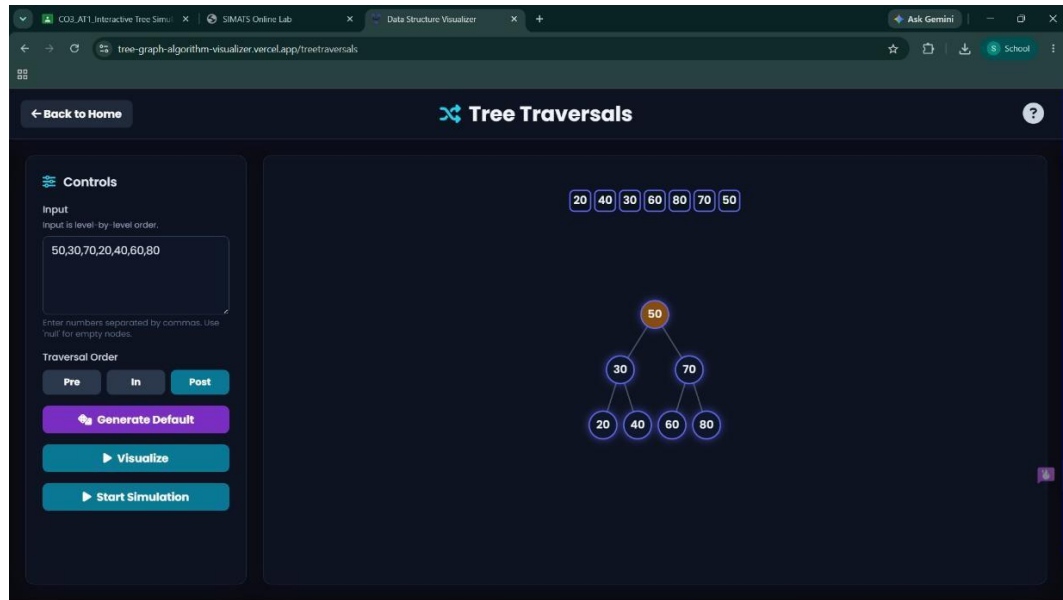
a. Inorder



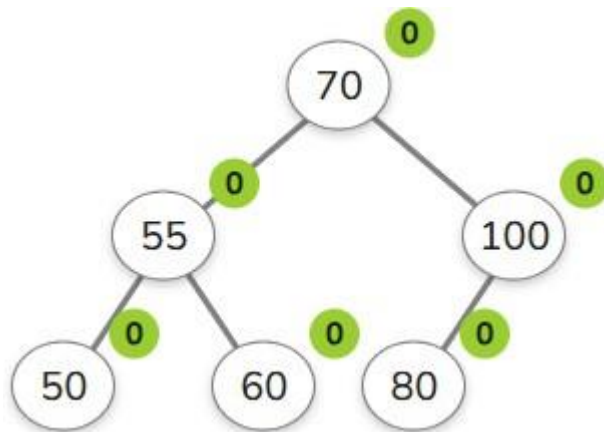
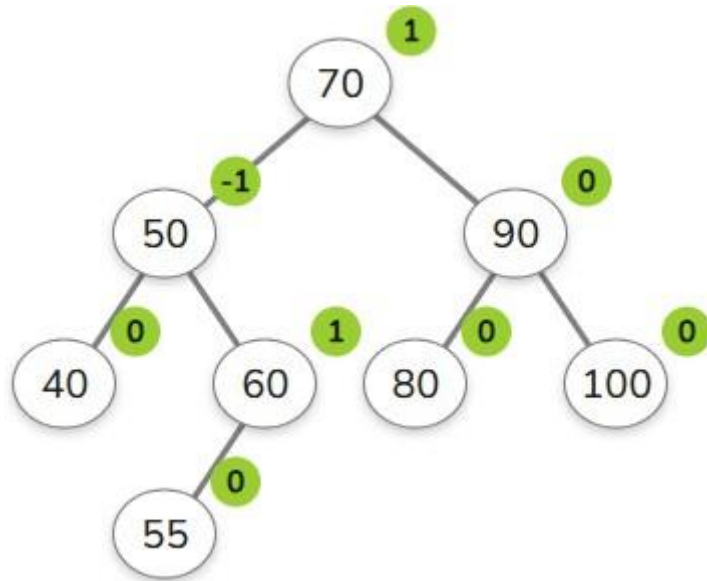
b. Preorder



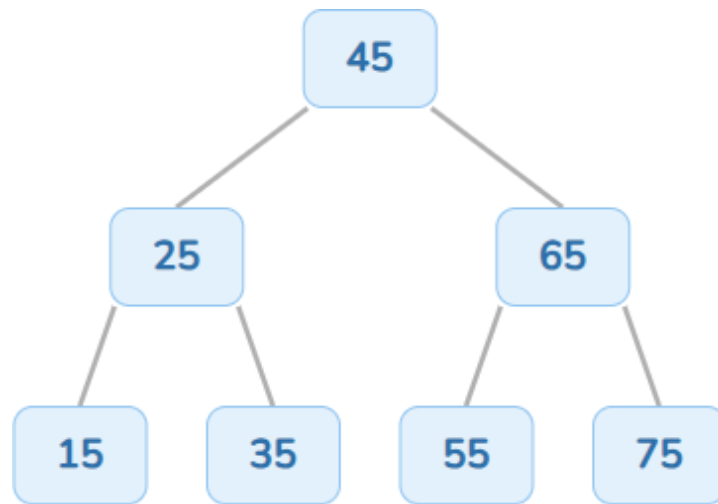
c. Postorder



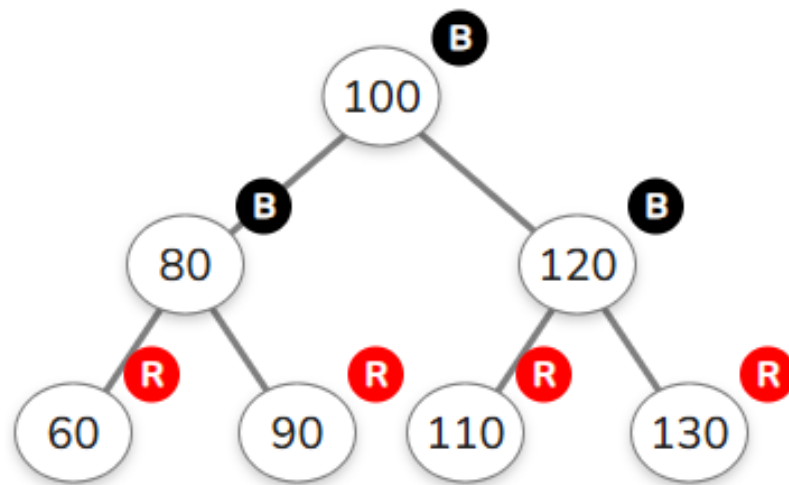
3. Insert 70, 50, 90, 40, 60, 80, 100, 55 into an AVL Tree. Delete 40 and 90. Analyze how the tree rebalances after each deletion.



4. Construct a B-Tree of order 3 by inserting 45, 25, 65, 15, 35, 55, 75



5. Insert 100,80,120,60,90,110,130 into a Red-Black Tree. Analyze their heights.



Insertion	Tree Height
100	0
80	1
120	1
60	2
90	2
110	2
130	2